SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code: CSA1006

Course Title: Software Engineering

CO Assessed: CO1 – Apply advanced methodologies and agile project management techniques to manage software projects effectively

Assessment: Assessment Tool 1 – Case Study Analysis (Weightage 20%, Total Marks 25, Duration 60 minutes)

Student Name: B. Pavani

Register No: 192525248

Date: 13/10/2026

Allotted Problem Statement: No. 4 – Remote Patient Monitoring and Predictive Healthcare Platform (Healthcare)

Code of Conduct

I, B. Pavani (192525248), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: B. Pavani

---

1. Understanding the Problem Statement

Many patients with long-term conditions such as hypertension, diabetes, heart disease and respiratory illness, as well as patients recovering after discharge, spend most of their time at home and are seen by a doctor only during occasional appointments. Between two visits, a patient's heart rate, oxygen saturation, blood pressure or temperature can drift into a dangerous range without anyone noticing. By the time the patient feels unwell enough to call or travel to hospital, the condition has often become an emergency. At the same time, hospitals face growing patient numbers and cannot place a nurse beside every patient. The problem is therefore to build a software platform that lets the care team observe patients continuously from a distance, recognise worsening health early, and act before a complication develops.

In my own words, the proposed system must collect vital signs from home devices and wearables, store them securely, show them to doctors and nurses on a live dashboard, calculate how risky each patient's current condition is, raise alerts when the risk rises, and give patients a simple application through which they can see their own readings, receive reminders and contact their doctor.

The objectives of the proposed system follow from this. The first objective is continuous, real-time monitoring of heart rate, SpO2, blood pressure, temperature and respiratory rate. The second is early detection of deterioration through a risk score that predicts which patients need attention

first. The third is timely notification of the right clinician when a patient crosses a threshold. The fourth is to keep patients involved in their own care through readings, reminders and easy communication. The fifth is to reduce unnecessary hospital visits and readmissions while protecting patient privacy through role-based, secure access to health data.

The scope of the problem covers the patient-facing application, the clinician web dashboard, the data collection and storage layer, the alert engine and the risk prediction module. Billing, pharmacy management, insurance claims, full electronic health record management and the design of the medical devices themselves lie outside the scope, although the system should be able to integrate with hospital records and accept readings from standard devices. The system is a decision-support tool, which means the final clinical judgement always remains with the doctor.

2. Key Stakeholders and Challenges

The first and most important stakeholder is the patient, who wears or uses the monitoring devices, records readings, follows medication reminders and contacts the doctor when needed. Patients are responsible for using the devices correctly and honestly. Doctors are the second group. They review the dashboard, interpret trends, decide on treatment changes and respond to high-risk alerts, and they carry the final clinical responsibility for the patient. Nurses form the third group and handle the day-to-day work: they watch the dashboard, call or visit patients whose readings look abnormal, verify measurements and escalate serious cases to the doctor. Family members and caregivers may receive notifications and help elderly patients use the application.

The hospital administration and the IT team are also stakeholders. Administrators decide the monitoring policy, manage staff accounts and roles and are accountable for regulatory compliance, while the IT team deploys, secures, monitors and maintains the platform and its integration with existing hospital systems. Device manufacturers supply the sensors and wearables whose data formats the platform must understand. Finally, regulators and data-protection authorities set the rules for handling sensitive health data, and the platform must follow them.

Each of these groups faces its own challenges. Patients, especially older ones, may find technology confusing, may forget to take readings and may worry about the privacy of their data. Doctors and nurses face information overload: if the dashboard shows every reading with equal weight, important changes are lost among normal values, and too many false alarms lead to alert fatigue. Clinicians also need trust in the numbers, which means that device errors and missing readings must be visible. Administrators struggle with the cost of devices and infrastructure and with integrating the platform with legacy hospital systems. The IT team must guarantee that the system stays available around the clock, because a monitoring platform that is down at night defeats its purpose, and it must protect the data from breaches. Across all groups there is the common challenge of unreliable home internet connections, which can delay or lose readings.

3. Analysis of the Current Approach

In the existing process, a patient with a chronic condition is asked to visit the clinic every few weeks or months. During the visit a nurse measures the vital signs and the doctor reviews the

results and adjusts the treatment. Between visits, some patients measure their blood pressure or sugar at home with personal devices and write the values in a notebook or remember them for the next appointment. If a patient feels very unwell, they telephone the hospital or go to the emergency department. Discharged patients are sometimes followed up by a phone call from a nurse, but this depends on staff availability.

This approach has genuine strengths. It is simple and familiar, requires no new technology, and the face-to-face examination allows a doctor to notice things that a device cannot measure. It is also inexpensive to start and does not raise the data-security concerns of a connected platform.

Its limitations are, however, serious. Monitoring is periodic rather than continuous, so deterioration between visits goes undetected. Readings kept in notebooks are incomplete, easy to forget and impossible for the doctor to analyse as a trend. There is no automatic alert, so the response starts only after symptoms are severe. Doctors have no way to rank patients by urgency, so a stable patient and a patient at risk wait in the same queue. Frequent travel to the hospital is tiring and costly for elderly and rural patients, and it increases the load on outpatient departments. The gaps that must be addressed are therefore the lack of continuous data, the lack of automated detection and prioritisation, the absence of timely alerts, the weak communication between visits and the lack of a shared, secure record of the patient's readings.

4. Proposed Software Solution Using Software Engineering Principles

The proposed solution is a platform named CareWatch, made up of a patient application and a clinician dashboard connected through a backend service. It is organised into several functional modules. The user management and authentication module registers doctors, nurses and patients and controls what each role may see. The vital signs collection module receives readings from home devices or from the patient's manual entry and stores them with a timestamp. The real-time monitoring module displays the latest readings and live trend charts on the dashboard. The risk prediction module converts the readings into a score between zero and one hundred and a risk level of Low, Medium or High. The alert engine raises a notification whenever a patient's risk level rises, and the patient communication module offers reminders, a measure-now option and a quick way to contact the doctor. A reporting module summarises each patient's history for review, and an administration module manages users, thresholds and integrations.

The functional requirements of the system are that it shall authenticate users by role, receive and store vital signs, display current values and trends, calculate a risk score for every patient, sort patients so that the highest risk appears first, generate and record alerts, send reminders to patients, let a patient trigger a new reading and contact the doctor, and produce history reports. The non-functional requirements are equally important. The system should update readings within a few seconds, remain available at all times, handle growth from a few patients to thousands, encrypt data in transit and at rest, keep an audit trail of who viewed which record, comply with health-data privacy rules, work on phones and desktops, and be simple enough for an elderly patient to use without training.

Software engineering principles are applied throughout the design. Modularity is achieved by separating the system into independent modules for authentication, data collection, risk scoring, alerts and presentation, each with a clear interface, so that a change in one module does not disturb the others. Scalability is supported by a stateless service architecture with a message queue between the data collection and processing layers, which allows more servers to be added as the number of monitored patients grows. Maintainability comes from this modular structure together with readable code, documentation and automated tests; in the prototype the risk rules are kept in one place so that a clinician-approved threshold can be changed without touching the user interface. Reliability is addressed by storing readings durably, retrying failed transmissions, buffering data on the device during network outages and running the services redundantly. Usability is built into the interface through colour coding, risk-based sorting, large readable tiles and a simple phone layout with only two main buttons. Security is incorporated through role-based access control, strong authentication, encrypted communication, encrypted storage and audit logs, because health data is among the most sensitive information a system can hold.

5. Justification of the Chosen Methodology

For this project I select the Agile approach using the Scrum framework. Scrum organises the work into short sprints of about two weeks, each producing a working increment that is demonstrated to the stakeholders, with a product backlog that the product owner can reprioritise after every sprint.

Scrum is suitable for this system for several reasons. The requirements of a healthcare platform are not fully known at the start, because doctors and nurses discover what information they actually need only when they see a real dashboard. Agile allows the team to build the dashboard prototype early, collect clinician feedback and refine the layout, alert rules and thresholds in the following sprints, instead of discovering a mismatch after the whole system is complete, as can happen in the Waterfall model. The system also has distinct, separable modules, such as monitoring, alerts and the patient app, which fit naturally into sprint-sized backlog items and let the most valuable features, such as vital signs display and risk alerts, be delivered first. Frequent releases reduce risk, because errors in the risk scoring or integration are found early when they are cheap to correct. Regular sprint reviews with doctors, nurses and patients keep the product aligned with real clinical needs, while daily stand-ups and retrospectives keep the team coordinated and continuously improving.

Scrum also supports maintenance after the first release. The backlog continues to receive new requests, such as support for new devices or a trained prediction model, and these are planned in later sprints. Combined with continuous integration and automated testing, this allows the platform to evolve safely. Because the system is safety-related, the Agile process would be supplemented with thorough documentation, clinical validation of the scoring rules and a strict security review before every release.

6. Expected Outcomes and Limitations

The expected benefits of the solution are considerable. Clinicians gain continuous visibility of their patients and can identify those at greatest risk within seconds of opening the dashboard, which supports earlier intervention and should reduce emergency admissions and readmissions. Patients receive reassurance and reminders, avoid unnecessary journeys and become more active participants in managing their health. Hospitals use nurse time more efficiently, because attention is directed to the patients who need it. Each earlier challenge is addressed directly: continuous readings replace periodic checks, the risk score and sorting solve the problem of prioritisation, the alert engine removes the delay in noticing a problem, the patient application improves communication between visits, and the central secure store replaces notebooks and scattered records.

There are also risks and limitations. The most important is clinical safety: an inaccurate threshold or a faulty sensor may generate a false alarm or, worse, miss a real emergency, so the output must be treated as decision support and not as a diagnosis. Too many false alerts may cause alert fatigue. Implementation challenges include the cost of devices, integration with existing hospital systems, staff training, resistance to change and patients who have poor internet access or limited technical skills. Data breaches and non-compliance with privacy rules are serious risks that demand constant attention. The prototype built for this assignment has its own limitations: it uses simulated data instead of real device readings, its login is handled only in the browser with demo accounts, its risk score is a simple set of rules rather than a trained model, and it has no real database or server.

Several future enhancements are possible. The platform can be connected to real wearables and medical devices through standard protocols, and the rule-based score can be replaced by a machine learning model trained on historical patient data and validated clinically. A backend with a database and WebSocket-based live updates, push notifications and SMS to doctors and caregivers, patient history reports, an administration panel, video consultation and integration with the hospital's electronic health records would turn the prototype into a complete production system.

7. Implementation of the Proposed System (Frontend Prototype)

To demonstrate the proposed solution, a working frontend prototype of CareWatch was developed. It is written in HTML, CSS and JavaScript in a single file and opens in a browser such as Chrome or Edge. It has two interfaces: a web dashboard used by doctors and nurses, and a mobile-style application used by patients. The charts are drawn as inline SVG using JavaScript, and the vital signs are simulated and refreshed every three seconds with the setInterval function so that the live behaviour of the real system can be demonstrated. The dashboard layout is designed for large screens, while the patient application uses a narrow phone layout. The following screenshots show the implementation.

7.1 Login Screen

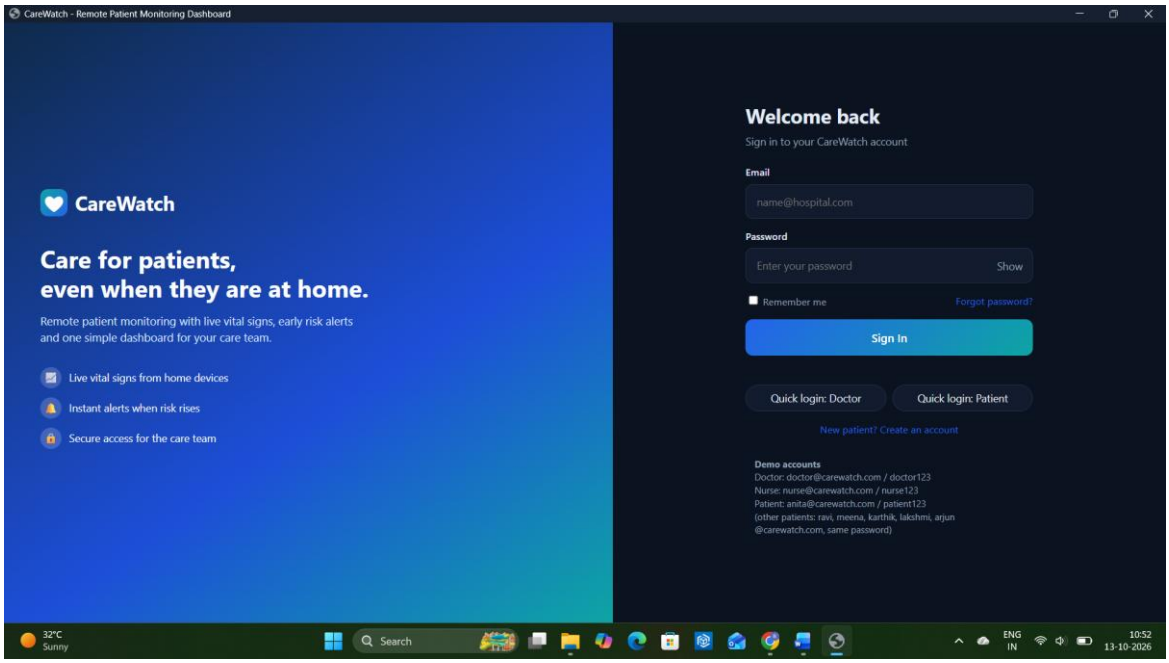


Figure 1: CareWatch login screen with role-based sign-in

Figure 1 shows the entry point of the system. The left side introduces the platform and its three main promises, which are live vital signs from home devices, instant alerts when risk rises and secure access for the care team. The right side contains the sign-in form with email and password fields, a show-password option, a remember-me option and a forgot-password link. For quick demonstration there are quick-login buttons for a doctor and a patient, a link for a new patient to create an account, and a list of demo accounts for the doctor, the nurse and the patients. After signing in, each user is taken to the interface that matches their role, which illustrates the role-based access required by the security principle.

7.2 Patient Application

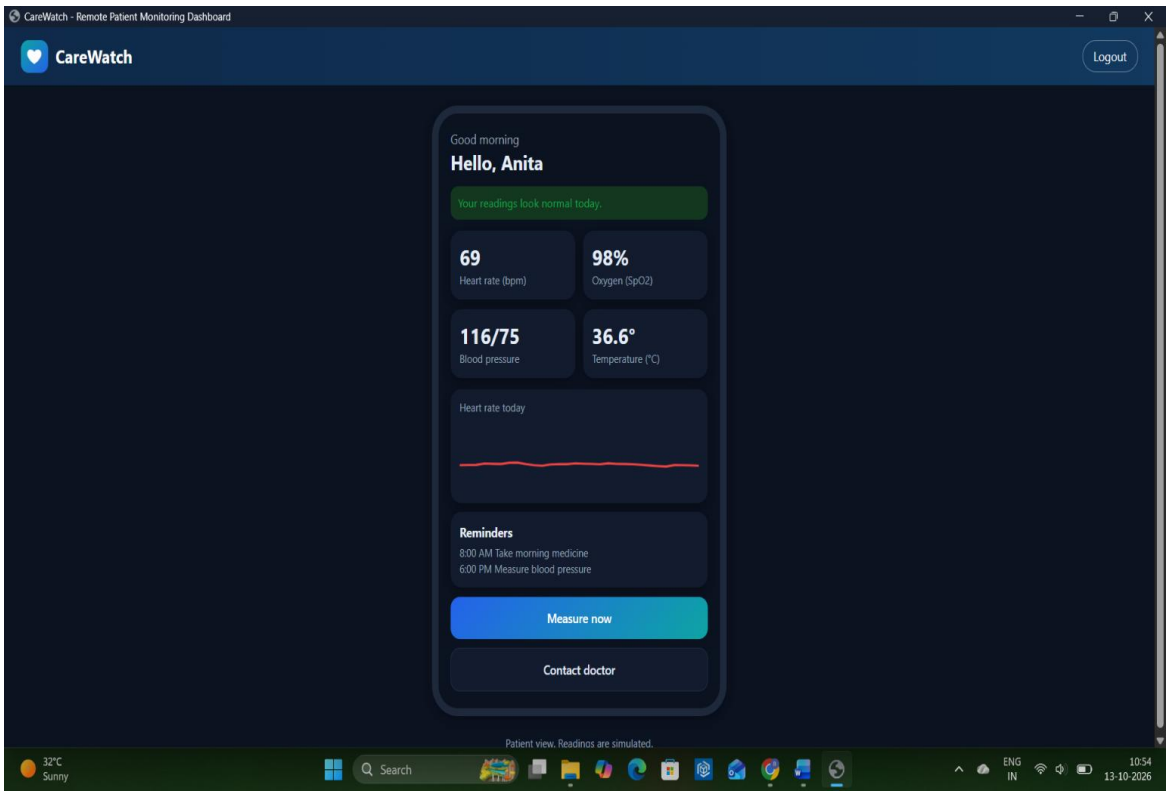


Figure 2: Patient application showing today's readings, trend chart and reminders

Figure 2 shows the patient application after a patient named Anita signs in. It greets the patient and displays a status message in a coloured box, which is green when the readings look normal. Four tiles show the heart rate of 69 beats per minute, oxygen saturation of 98 percent, blood pressure of 116/75 and temperature of 36.6 degrees Celsius. Below them a small line chart shows the heart rate during the day, followed by a reminders box listing the morning medicine at 8:00 AM and the blood pressure measurement at 6:00 PM. At the bottom, the Measure now button records a new reading and sends it to the doctor, and the Contact doctor button lets the patient reach the doctor quickly. The simple layout, large numbers and only two actions make the application usable for elderly patients.

7.3 Doctor Dashboard

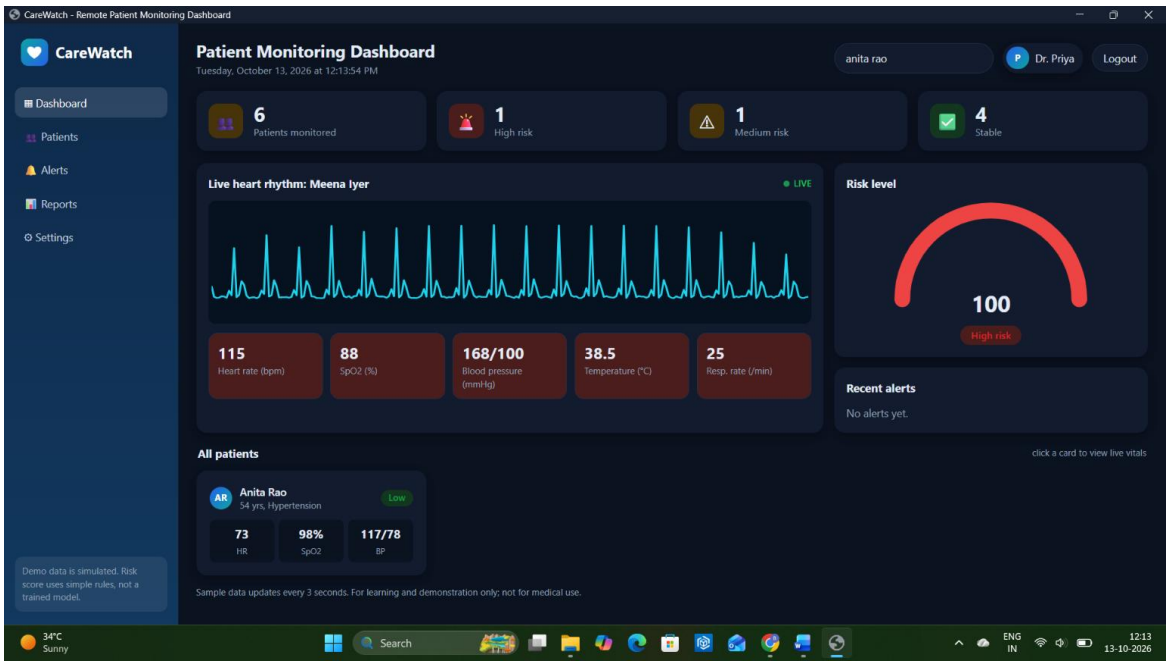


Figure 3: Doctor dashboard with summary cards, live heart rhythm and risk gauge

Figure 3 shows the dashboard after a doctor, Dr. Priya, signs in. A navigation menu on the left provides the Dashboard, Patients, Alerts, Reports and Settings sections, and a search box at the top finds a patient by name. Four summary cards give the number of patients monitored, which is six, and how many are at high risk, medium risk and stable. The main panel shows the live heart rhythm of the selected patient, here Meena Iyer, drawn as a continuously moving trace, with tiles for heart rate of 115, SpO2 of 88 percent, blood pressure of 168/100, temperature of 38.5 degrees and respiratory rate of 25. Because these values are outside their normal ranges they are shaded red. The risk gauge on the right shows a score of 100 with the label High risk, and a recent alerts panel lists the latest notifications. Below, the All patients section presents a card for each patient with the name, age, condition, current vital signs and a risk badge, and clicking a card loads that patient's live vitals in the main panel.

7.4 Nurse View and Alerts



Figure 4: Nurse dashboard showing a low-risk patient and the recent alerts panel

Figure 4 shows the same dashboard for a nurse, Nurse Kavya. The selected patient is Anita Rao, whose heart rate of 72, SpO2 of 98 percent, blood pressure of 118/76, temperature of 36.7 degrees and respiratory rate of 16 are all normal, so the tiles are green and the risk gauge shows a score of zero with the label Low risk. The recent alerts panel on the right records messages generated by the alert engine; in this screenshot it shows that the patient Lakshmi N moved to Medium risk, with her heart rate and SpO2 values included in the message, at the time the change occurred. This demonstrates that a nurse can watch the whole ward at a glance and is told automatically when someone needs attention, without searching through individual records.

7.5 How the Risk Score Works

The prototype uses a simple rule-based risk score that is easy for clinicians to understand and adjust. Points are added whenever a vital sign moves outside its normal range, and the total, limited to a maximum of one hundred, decides the risk level. For heart rate, 35 points are added if it is above 110 or below 50, and 15 points if it is above 100 or below 55. For oxygen saturation, 40 points are added if it is below 90 percent and 20 points if it is below 94 percent. For systolic blood pressure, 25 points are added above 160 and 12 points above 140. For temperature, 20 points are added above 38 degrees Celsius and 8 points above 37.5 degrees. A patient is classed as High risk if the score is 60 or more, Medium risk if it is 25 or more, and Low risk otherwise. For example, the readings of Meena Iyer in Figure 3 earn points for heart rate, SpO2, blood pressure and temperature together, so her score reaches the maximum of 100 and she is shown as High risk, whereas the normal readings of Anita Rao in Figure 4 earn no points and give a score of zero.

7.6 Mapping of the Prototype to the Proposed System

The prototype demonstrates the main modules described in Section 4. Real-time vital monitoring appears in the vital sign tiles and the live trend and heart rhythm charts. The alert engine appears as the recent alerts panel. Risk prediction appears as the risk badge, the gauge and the score for every patient. The patient application module appears in the phone-style interface with readings, reminders and the Measure now and Contact doctor buttons. Usability is shown through colour coding, risk-based sorting of patients and the simple layout, and security and role handling are illustrated by the login screen and the separate doctor, nurse and patient views. The prototype uses simulated data and has no real backend, which is stated clearly as a limitation in Section 6.